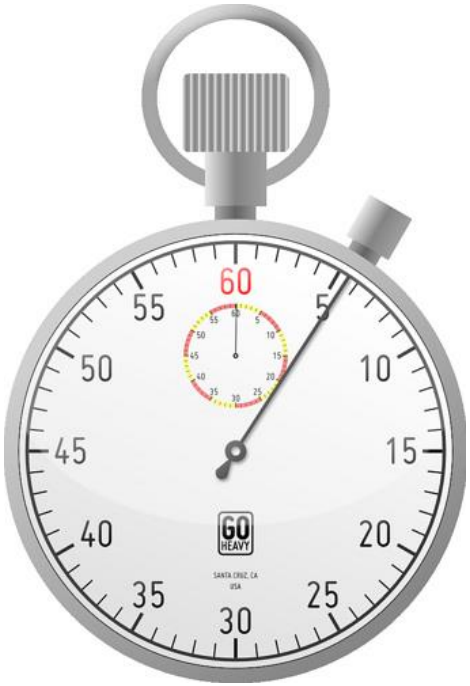
A BASIC introduction to Distributed Ledger ("Blockchain") Technology

AGENDA

Distributed Ledger / Blockchain Technology

In five minutes…



Gary Nuttall (Distlytics Ltd)
May 2016

What is it? …

Definition of *blockchain* in English:

blockchain

Pronunciation: /ˈblɒktʃeɪn/

NOUN

A digital ledger in which transactions made in bitcoin or another cryptocurrency are recorded chronologically and publicly:
'we can actually have a look at the blockchain and see evidence of what's going on'

+ More example sentences

Block chain (database) From Wikipedia, the free encyclopedia
Jump to: navigation, search

A **block chain** or **blockchain** is a distributed database, introduced in Bitcoin, that maintains a continuously-growing list of data records that each refer to previous items on this list and is thus hardened against tampering and revision.[1][disputed – discuss]

The initial and most widely known application of block chain technology is the public ledger of transactions for bitcoin,[2] which has been the inspiration for other cryptocurrencies[3] and distributed database designs.[4]

TOO COMPLICATED!

WIKIPEDIA
The Free Encyclopedia

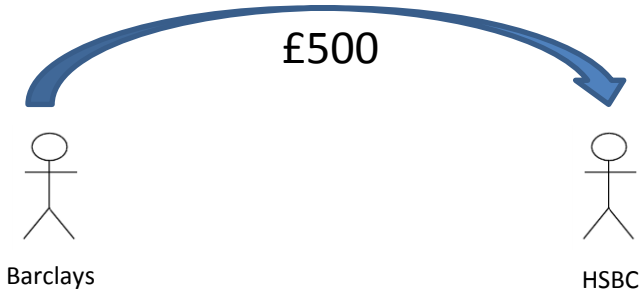
Let’s look instead at a process…

A basic transaction between two parties

Transferring value

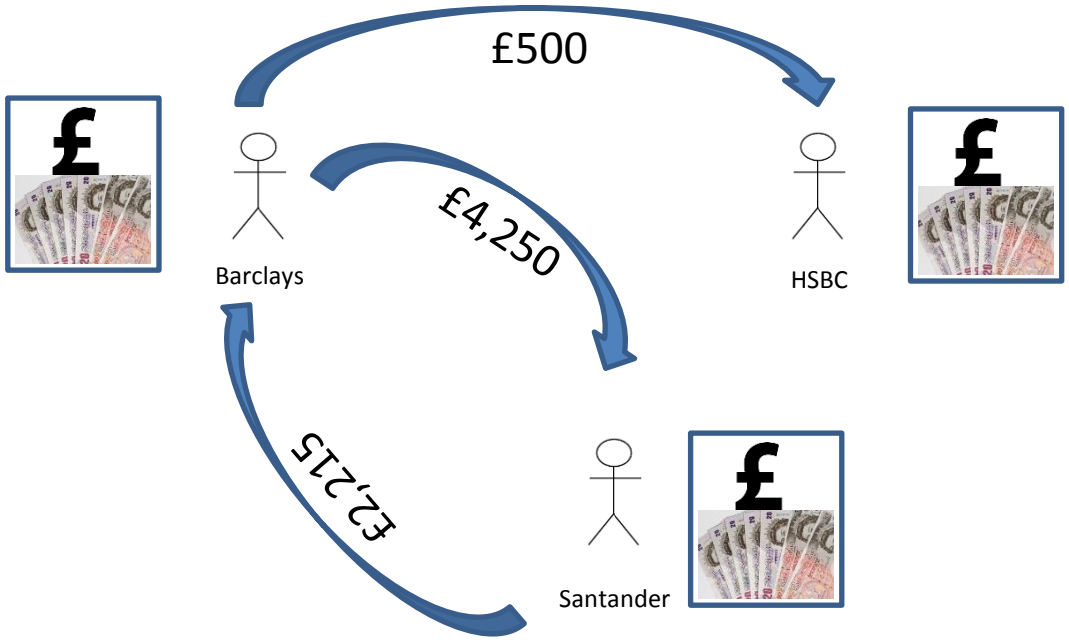
A **Ledger** is simply a journal of transactions

JOURNAL -ID	DATESTAMP	FROM	TO	CURRENCY	AMOUNT
1	01/01/2016 08:35	BARCLAYS	HSBC	GBP	500.00



…usually lots of transactions

BARCLAYS JOURNAL					
JOURNAL-ID	**DATESTAMP**	**FROM**	**TO**	**CURRENCY**	**AMOUNT**
1	01/01/2016 08:35	BARCLAYS	HSBC	GBP	500.00
2	01/01/2016 09:45	BARCLAYS	SANTANDER	GBP	4,250.00
3	01/01/2016 11:35	SANTANDER	BARCLAYS	GBP	2,215.00



Everyone has a copy of their own ledger

BARCLAYS JOURNAL					
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HSBC JOURNAL					
JOURNAL-ID	DATESTAMP	FROM	TO	CURRENCY	AMOUNT
1	01/01/2016 08:35	BARCLAYS	HSBC	GBP	500.00
4	01/01/2016 13:35	HSBC	SANTANDER	GBP	105.00

SANTANDER JOURNAL					
JOURNAL-ID	DATESTAMP	FROM	TO	CURRENCY	AMOUNT
2	01/01/2016 09:45	BARCLAYS	SANTANDER	GBP	4,250.00
3	01/01/2016 11:35	SANTANDER	BARCLAYS	GBP	2,215.00

PROBLEMS!

RECONCILIATION – Need to check that <u>every</u> record of <u>every</u> transaction has been copied to each other's systems correctly.

AUDIT – Need to prove that the system works consistently

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AUDIT – Need to prove that the system works consistently

COSTS

- **Processing overhead (needs computing power)**
- **Reconciliation systems need to be designed, developed, tested & supported**
- **Excel addicts writing macros, functions & stuff that goes wrong**
- **Auditing overhead – cost of auditors, audits, etc.**
- **Data Quality issues – End up "working around" problems**

TIME

- **Need to wait for reconciliations to be executed and verified**
- **Audits are after the event and aren't preventative**

SOLUTION!

Merge all the Ledgers into one…

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A Mutual Ledger…Less reconciliation

BLOCK - CHAIN

Put the data into blocks and chain them together

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4	01/01/2016 13:35	HSBC	SANTANDER	GBP	105.00

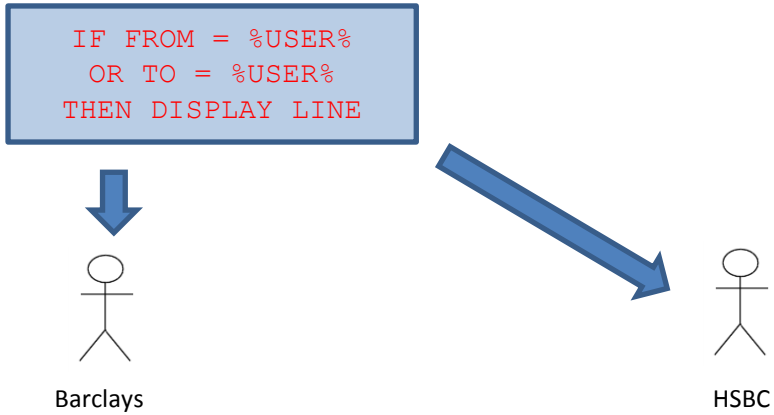
BLOCK 1				BLOCK 2				BLOCK 3		
DATA	PREV HASH	CURRENT HASH		DATA	PREV HASH	CURRENT HASH		DATA	PREV HASH	CURRENT HASH
1	1011	1100		1	1100	1101		1	1101	1111

…now very difficult to change data in a prior block

Which decreases Auditing overhead

Put crypto-security onto the Ledger

JOURNAL-ID	DATESTAMP	FROM	TO	CURRENCY	AMOUNT	HASH	BLOCK	BLOCK HASH	START BLOCK	START HASH
1	01/01/2016 08:35	BARCLAYS	HSBC	GBP	500.00	1111			0	110111
2	01/01/2016 09:45	BARCLAYS	SANTANDER	GBP	4,250.00	101				
3	01/01/2016 11:35	SANTANDER	BARCLAYS	GBP	2,215.00	100011				
4	01/01/2016 13:35	HSBC	SANTANDER	GBP	105.00	101101	1	1111000		



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1	01/01/2016 08:35	**BARCLAYS**	HSBC	GBP	500.00
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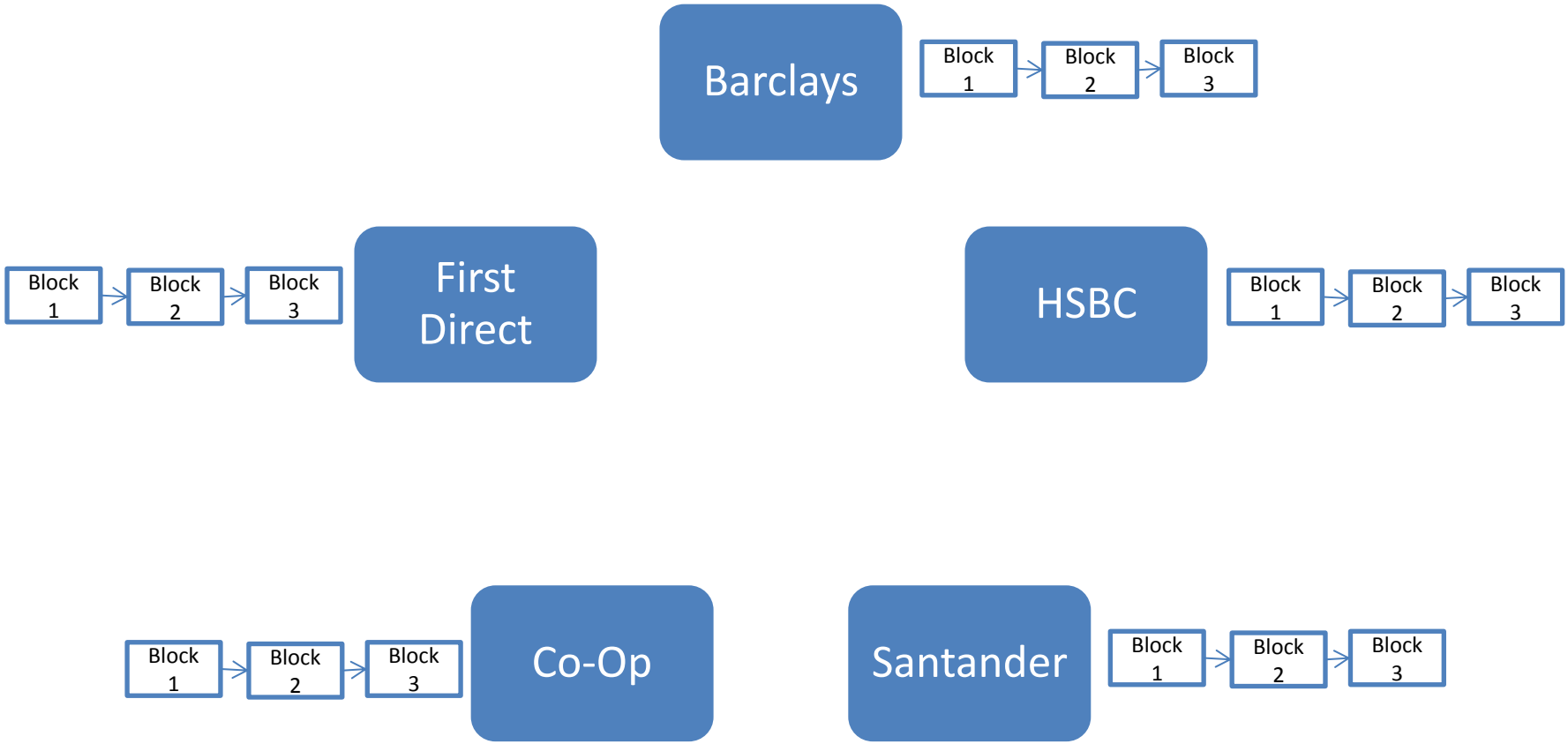
...now users can only access their own data

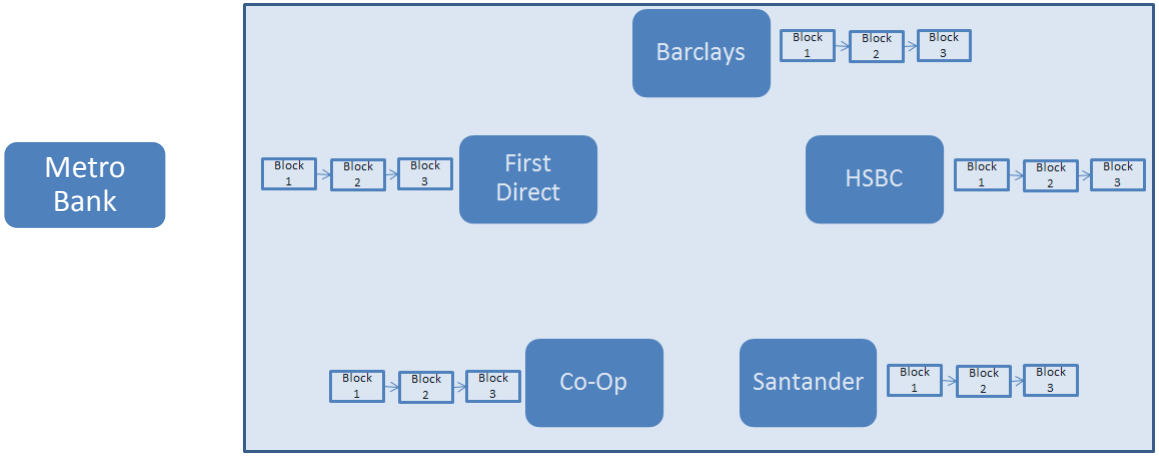
Which decreases Security Auditing overhead

Distribute a copy of the ledger to everyone



…And you have a <u>Mutual Distributed Ledger</u>

Restrict access only to members…



…Giving a Private, Permissioned Ledger

So, back to the definition….

It's a <u>write-only</u> database
That <u>everyone</u> has an identical copy of

With all entries <u>timestamped</u>
And the data is cryptographically <u>secured</u>

So, back to the definition….

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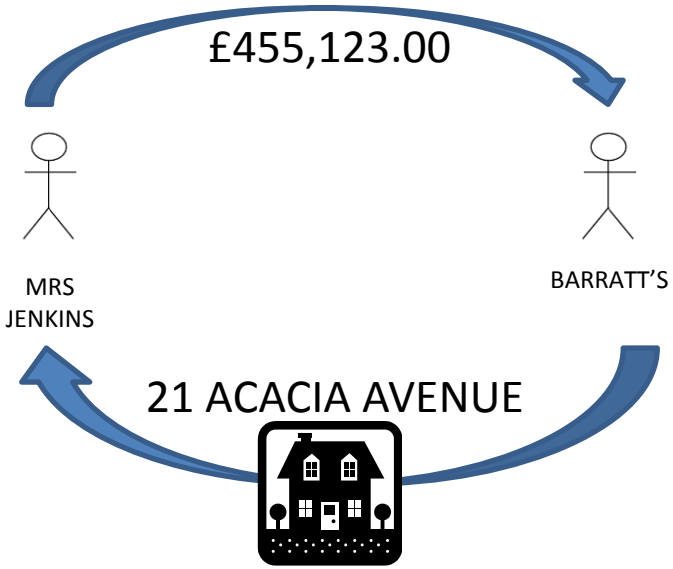
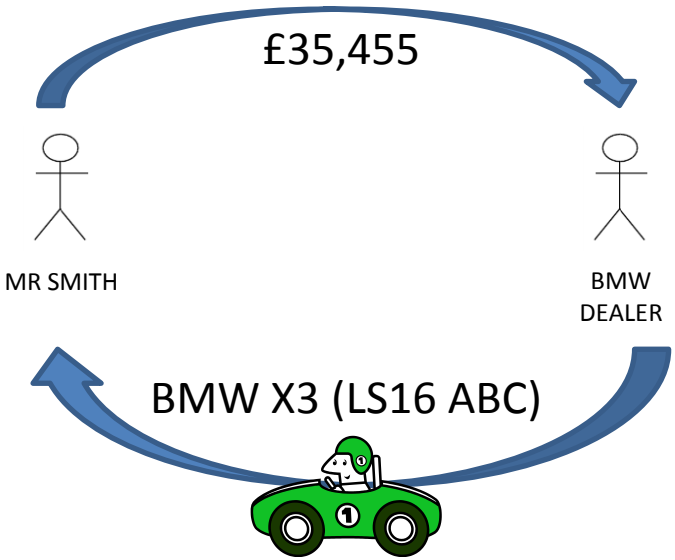
Which means:

- **A complete history of all transactions - great audit trail**
- **Everyone has a copy of the same thing - No need for reconciliation**
- **It's highly distributed – Makes it cyber-resistant**
- **Data is cryptographically secured – overcomes security issues**

But, there’s more!

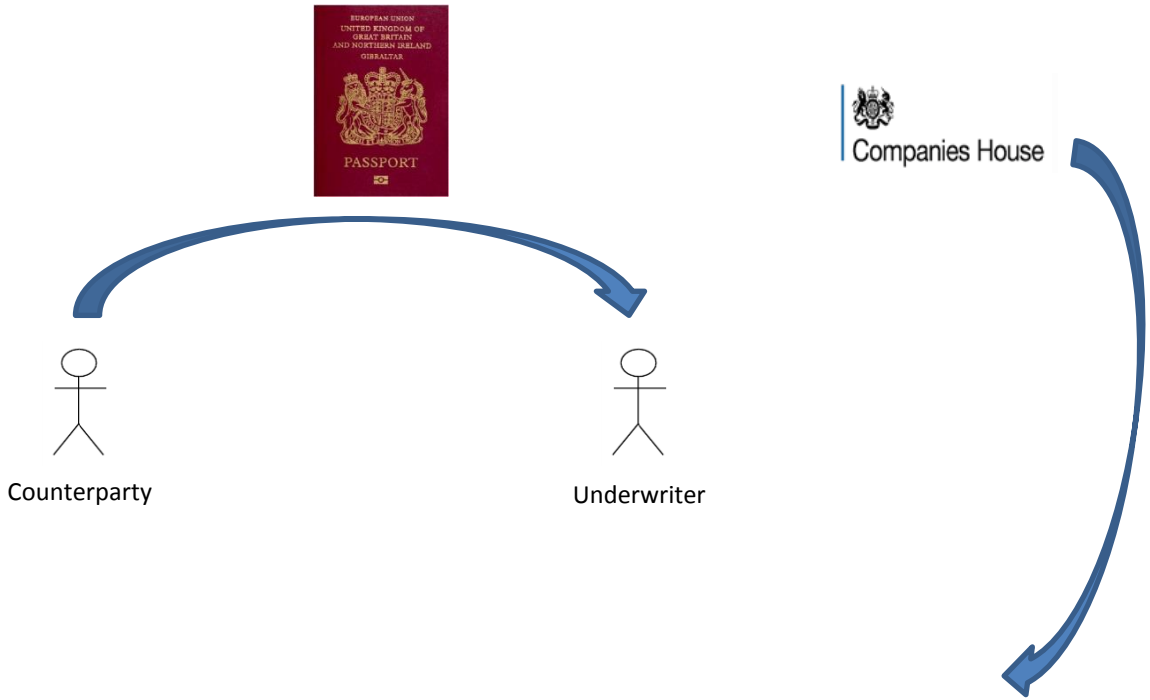
Other things could go onto a ledger…such as assets





Journal ID	Datestamp	FROM	TO	UNIT	IDENTIFIER
1	01/01/2016 11:45:00	MRS JENKINS	BARRATT'S	GBP	455,123.00
2	01/01/2016 11:45:01	BARRATT'S	MRS JENKINS	HOUSE	21 ACACIA AVENUE
3	02/01/2016 10:35:00	MR SMITH	BMW DEALER	GBP	35,455.00
4	02/01/2016 10:35:01	BMW DEALER	MR SMITH	CAR	BMW X3 (LS16 ABC)

Links to other things….such as documents



Journal ID	Datestamp	TYPE	REFERENCE	CLASS	IDENTIFIER
1	01/01/2016 11:45:00	KYC	PASSPORT	PDF	ABC123
2	01/01/2016 11:45:01	KYC	DIR CHECK	PDF	ABC124
3	02/01/2016 10:35:00	CLAIM	FNOL	EMAIL	DEF124
4	02/01/2016 10:35:01	CLAIM	ASSESS	WORD	DOC121

Then there's Smart Contracts…

Journal ID	Datestamp	FROM	TO	UNIT	IDENTIFIER
1	01/01/2016 11:45:00	ATRIUM	LUFTHANSA	EUR	POLICY123
2	01/01/2016 11:45:01	BEAZLEY	FARMER MCGREGOR	GBP	POLICY234

POLICY123	IF (POLICY_IS_ACTIVE AND PREMIUM_PAID AND CLAIM_CONDITION_MET THEN PAY_CLAIM

POLICY234	IF (POLICY_IS_ACTIVE AND PREMIUM_PAID AND CLAIM_CONDITION_MET THEN PAY_CLAIM

Connect with data sources, providing unquivocable evidence of…

Weather conditions (Temperature, precipitation)

Flight Delays

Automating Claim Notification process, thereby

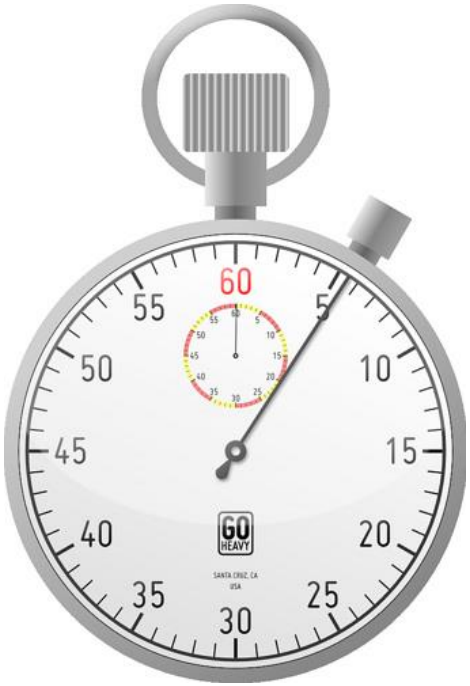
Decreasing Cost of Claim Management

Speeding up Claim

Improving Customer Experience

A few caveats…

THANKS!



USEFUL RESOURCES
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